

AAA Digital Platform

Technical Setup Guide

Developer and DevOps reference for setup and deployment

This document is the authoritative technical reference for the AAA Digital Platform. It covers the full technology stack, environment configuration, database setup, deployment pipeline, and troubleshooting guidance for developers and DevOps engineers.

Version 1.0 | May 2026 | Ambitious about Autism

1. Architecture Overview

Layer	Technology	Hosted at
Frontend & API	Next.js 14 App Router, TypeScript	Vercel (asd-training-app-v2)
Database	PostgreSQL, Prisma ORM	Neon (pooler port 6543)
Authentication	NextAuth v4 JWT sessions, TOTP MFA	In-app
File Storage	Vercel Blob	Vercel
AI Features	Vercel AI Gateway → Gemini / Claude / GPT	Vercel
Text-to-Speech	ElevenLabs API (Lily voice, cached to Blob)	ElevenLabs
Email	Resend API (transactional)	Resend
Payments	Stripe (ENABLE_PAYMENTS=false in prod)	Stripe

Route groups: (auth) → public login/register/reset; (dashboard) → leaf roles; (org-admin) → ORG_ADMIN; (super-admin) → SUPER_ADMIN; (mfa) → MFA flow; (change-password) → forced change.

Auth flow: credentials/SSO → JWT callback extends token (id, role, organisationId, mustChangePassword, totpEnabled, mfaPending, hasPassword, effectivePrograms, charityPermissions, cvBuilderEnabled, careersAdvisorEnabled, isParentOrg) → session callback → middleware checks role / MFA / password.

2. Prerequisites

Tool	Version	Purpose
Node.js	18+ (20 LTS recommended)	Runtime
npm	9+	Package management
Git	Any recent version	Source control
Stripe CLI	Latest	Local webhook forwarding (optional)
Playwright	Installed via npm	E2E testing

External service accounts required: Vercel, Neon, GitHub, Resend. Optional: ElevenLabs (TTS), Google Cloud (Google SSO), Azure AD (Microsoft SSO).

3. Repository Setup

```
git clone https://github.com/Hodgy007/asd-training-app.git
cd asd-training-app
npm install
```

Script	Command	Purpose
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dev	npm run dev	Start dev server on localhost:3000
build	npm run build	prisma generate + vitest run + next build
test	npm run test	Run Vitest unit tests
test:e2e	npm run test:e2e	Run Playwright E2E tests
prisma:push	npm run prisma:push	Push schema to database
prisma:studio	npm run prisma:studio	Open Prisma Studio
prisma:seed	npm run prisma:seed	Seed demo users
prisma:generate	npm run prisma:generate	Regenerate Prisma client after schema changes

4. Key Directory Structure

Path	Purpose
app/(auth)/	Login, register, forgot-password, reset-password (public)
app/(dashboard)/	Leaf-role pages (dashboard, training, cv-builder, jobs, etc.)
app/(org-admin)/	Org admin pages (/admin/*)
app/(super-admin)/	Super admin pages (/super-admin/*)
app/(mfa)/	MFA setup and verify pages
app/api/	All API route handlers
components/	Shared React components
lib/	Server-side utilities, AI runner, auth helpers, RBAC
prisma/	schema.prisma, seed.ts, migrations
types/	TypeScript type extensions (session, next-auth)
middleware.ts	Route protection, MFA enforcement, role redirects
public/	Static assets (logo, favicon)

5. Environment Variables

C R I T I C A L

DATABASE_URL must use the Neon POOLER URL (port 6543) in production. Using the direct URL (port 5432) will exhaust connection limits on serverless.

Variable	Required	Description
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DATABASE_URL	Yes	Neon pooler — port 6543 with ?pgbouncer=true
DIRECT_URL	Yes	Neon direct — port 5432, Prisma migrations only
NEXTAUTH_SECRET	Yes	32+ char random secret: openssl rand -base64 32
NEXTAUTH_URL	Yes	Production URL, no trailing slash
GEMINI_API_KEY / AI_GATEWAY_API_KEY	Yes	AI features via Vercel AI Gateway
RESEND_API_KEY	Yes	Transactional email
BLOB_READ_WRITE_TOKEN	Yes	Vercel Blob file storage
ELEVENLABS_API_KEY	Yes	Text-to-speech (Lily voice)
GOOGLE_CLIENT_ID	Optional	Disables Google SSO if absent
GOOGLE_CLIENT_SECRET	Optional	Google OAuth client secret
AZURE_AD_CLIENT_ID	Optional	Disables Microsoft SSO if absent
AZURE_AD_CLIENT_SECRET	Optional	Azure AD OAuth client secret
AZURE_AD_TENANT_ID	Optional	Use "common" for all account types
DISABLE_MFA	Optional	Set true to skip MFA enforcement (dev only)
ENABLE_PAYMENTS	Optional	Set true to expose /courses and Stripe checkout
STRIPE_SECRET_KEY	Payments	Stripe secret key (sk_test_... / sk_live_...)
STRIPE_PUBLISHABLE_KEY	Payments	Stripe publishable key
STRIPE_WEBHOOK_SECRET	Payments	Stripe webhook signing secret (whsec_...)

```
# Pull env vars from Vercel (dev branch):
npx vercel env pull .env.local

# Pull production env vars:
npx vercel env pull .env.production --environment production --yes
```

6. Database Setup (Neon)

Two Neon branches are in use: **main** (production, endpoint: ep-blue-thunder-a88kb0cy) and **dev** (development, endpoint: ep-lucky-cherry-a8toqlw5). Local dev pulls the dev branch env vars via `npx vercel env pull .env.local`.

```
# Push schema changes to dev database:
npm run prisma:push

# Push schema to PRODUCTION:
npx dotenv-cli -e .env.production -- npx prisma db push

# Seed demo users:
npm run prisma:seed

# Seed training content:
npx tsx prisma/seed-training-content.ts
```

WARN

Always push schema to BOTH dev AND production when adding columns that are queried at runtime. Production will crash if it references a missing column.

Prisma accessor gotchas: The CV model accessor is `prisma.cv` (not `prisma.cv`). Virtual classroom sessions use `prisma.classSession` (not `prisma.session` — that is the NextAuth Session table).

7. Local Development

```
npm run dev
# Server starts on http://localhost:3000

# Forward Stripe webhooks (separate terminal):
stripe listen --forward-to localhost:3000/api/stripe/webhook
```

Demo credentials (seeded by `npm run prisma:seed`):

Role	Email	Password
SUPER_ADMIN	admin@asdawareness.org.uk	admin123
EMPLOYEE	demo@example.com	demo123

WARN

The dev Neon branch may be paused after inactivity. If login fails with a database connection error, wake the branch up in the Neon console.

8. Testing

```
npm run test # Vitest unit tests
npm run test:e2e # Playwright E2E tests
npm run build # Build (runs tests – failing test blocks deploy)
```

Test type	Location	What is tested
Vitest unit	lib/__tests__/ 	RBAC helpers, permission counts, job visibility logic
Playwright E2E	playwright-report/ 	Full browser flows — login, training, admin
Build gate	npm run build	Vitest runs before next build — failure = blocked deploy

Do NOT use isomorphic-dompurify or jsdom at runtime — they break on Vercel serverless. jsdom is a devDependency only (Vitest env). Use sanitize-html for HTML sanitisation.

9. Deployment (Vercel)

Vercel GitHub integration auto-deploys on push to **main**. Preview deployments are created for all branches.

```
# Deploy to production via git (preferred):
git push origin main

# Manual deploy (only if no git push was made):
npx vercel deploy --prod
```

Never run `vercel deploy --prod` after a git push — Vercel auto-deploys from GitHub and a manual deploy creates a duplicate build.

Step	Action
1	Vercel detects push to main
2	Runs build command: prisma generate && cross-env NODE_ENV=test vitest run && next build
3	Deploys to production URL: https://asd-training-app-v2.vercel.app
4	Preview URL available for branch deploys

10. Content Security Policy

CSP is configured in next.config.js headers. Key directives that require explicit values:

Directive	Required values	Reason
connect-src	vercel.com, blob.vercel-storage.com, *.public.blob.vercel-storage.com	Blob client-direct uploads
media-src	blob:	TTS audio playback
frame-src	blob:	SCORM iframe content
script-src	self, unsafe-eval (dev only)	Next.js HMR
font-src	fonts.gstatic.com	Google Fonts

SCORM iframe sandbox requires allow-same-origin. Removing it causes subresource requests to 401. Do not attempt to harden without a signed-URL scheme.

11. Key Architectural Decisions

NextAuth uses JWT sessions (not database sessions) with manual SSO account linking in the signIn callback. This avoids OAuthAccountNotLinked errors and simplifies the auth flow.

isomorphic-dompurify requires jsdom which breaks on Vercel serverless. sanitize-html is a pure Node.js HTML sanitiser with no browser dependencies.

The SCORM asset route (/api/scorm) buffers binary bodies before responding. Streaming corrupts media files — same issue as the TTS route.

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The /api/tts route streams MP3 bytes directly rather than redirecting to Blob. This allows magic-byte validation and prevents browser cache collisions.

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Supabase was IPv6-only — incompatible with Vercel Lambda which uses IPv4. Neon supports both. Migration was completed in May 2026.

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scorm-again handles both SCORM 1.2 (window.API) and SCORM 2004 (window.API_1484_11). Version 3 supports concurrent requests and improved CMI state management.

Prisma generates accessors from model names. The CV model is named "CV" (uppercase) so the accessor is prisma.cV. Similarly ClassSession avoids collision with NextAuth Session.

12. Troubleshooting

Symptom	Likely cause	Fix
All /api/auth/* return 500	Wrong DATABASE_URL (direct not pooler)	Use Neon pooler URL port 6543
Prisma client not found on Vercel	Client not regenerated before build	Build script must run prisma generate first
Login fails locally	Dev Neon branch paused	Wake branch in Neon console
Google SSO fails	Wrong redirect URI	Check Google Cloud Console → Credentials
Azure SSO fails	Wrong redirect URI or tenant	Check Azure Portal → App registrations
SCORM audio/video corrupted	Streaming not buffering	Ensure route buffers full body before res.send
TTS returns 200 but no audio	Non-MP3 response from ElevenLabs	Check ELEVENLABS_API_KEY; inspect magic bytes
Cookie banner blocks screenshots	CSP or cookie middleware	Accept cookies or set permissive CSP in test
Build fails — test error	Vitest runs in build	Fix failing unit test before deploying
MFA redirect loop	TOTP not enrolled	Log in as admin, go to /mfa-setup, complete setup